Bolt factory Machines A, B & C manufacture 20%, 30%, & 50% of the total of their output and 6%, 3%, & 2% are defective. A bolt is drawn at random and found to be defective. Find the Probabilities that it is manufactured from i) Machine - A ii) Machine - B iii) Machine - C

Sol^n:- let P(A), P(B), P(C) be the probability of the events that are manufacture by the machines A, B, C respectively then

$P(A) = 20\% = \frac{20}{100} = \frac{1}{5}$

$P(B) = 30\% = \frac{30}{100} = \frac{3}{10}$

$P(C) = 50\% = \frac{50}{100} = \frac{1}{2}$

let 'D' Denote the is defective them

$P(D/A) = 6\% = 6/100$

$P(D/B) = 3\% = 3/100$

$P(D/C) = 2\% = 2/100$

i) If both is defective then the probability that it is from A

i.e $P(A/D) = \dfrac{P(A).\ P(D/A)}{P(A)\cdot P(D/A) + P(B)\cdot P(D/B) + P(C)\cdot P(D/C)}$

$$= \frac{\frac{1}{5} \times \frac{6}{100}}{\frac{1}{5} \times \frac{6}{100} + \frac{3}{10} \times \frac{3}{100} + \frac{1}{2} \times \frac{2}{100}}$$

$$= \frac{\frac{6}{500}}{\frac{6}{500} + \frac{9}{1000} + \frac{1}{100}}$$

$$= \frac{\frac{6}{500}}{31/1000}$$

$$= \frac{6}{500} \times \frac{1000}{31}$$

$$= \frac{12}{31}$$

ii) Machine B:-

$$P[B/D] = \frac{P[B] \cdot P[D/B]}{P(A) \cdot P(D/A) + P(B) \cdot P(D/B) + P(C) \cdot P(D/C)}$$

$$= \frac{\frac{3}{10} \times \frac{3}{100}}{\frac{1}{5} \times \frac{6}{100} + \frac{3}{10} \times \frac{3}{100} + \frac{1}{2} \times \frac{2}{100}}$$

$$= \frac{\frac{9}{1000}}{\frac{31}{1000}}$$

$$P(B/D] = \frac{9}{31}$$

iii) Machine C:-

$$P[C/D] = \frac{P[C] \cdot P[D/C]}{P(A) \cdot P(D/A) + P(B) \cdot P(D/B) + P(C) \cdot P(D/C)}$$

$$= \frac{\frac{1}{2} \times \frac{2}{100}}{\frac{1}{5} \times \frac{6}{100} + \frac{3}{10} \times \frac{3}{100} + \frac{1}{2} \times \frac{2}{100}}$$

$$= \frac{\frac{1}{100}}{\frac{31}{1000}}$$

$$= \frac{1}{100} \times \frac{1000}{31}$$

$$= \frac{10}{31}$$

$$P[C/D] = \frac{10}{31}$$

random variable x is distributed at random between the values 0 and 1 so that its probability density function is $f(x) = kx^2(1-x^3)$ where k is a constant, then find i) k, ii) Mean iii) variance and iv) $P(0<x<0.5)$

Soln:- Given Probability density function $f(x) = kx^2(1-x^3)$

$$0 \le x \le 1$$

we have $\int_{-\infty}^{\infty} f(x)dx = 1$

$$\int_0^1 kx^2(1-x^3)\,dx = 1$$

$$k\int_0^1 (x^2 - x^5)\,dx = 1$$

$$k\left[\frac{x^3}{3} - \frac{x^6}{6}\right]_0^1 = 1$$

$$k\left[\frac{1}{3} - \frac{1}{6}\right] = 1$$

$$= k\left[\frac{1}{6}\right] = 1 \Rightarrow k = 6$$

ii) Mean $\mu = \int_{-\infty}^{\infty} x f(x)\,dx$

$$= \int_0^1 x \cdot kx^2(1-x^3)\,dx$$

$$= 6\int_0^1 (x^3 - x^6)\,dx$$

$$= 6\left[\frac{x^4}{4} - \frac{x^7}{7}\right]_0^1$$

$$= 6\left[\frac{1}{4} - \frac{1}{7}\right]$$

$$\text{Mean} = 6\left[\frac{7-4}{28}\right] = \frac{18}{28} = \frac{9}{14}$$

iii) Variance

$$\sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

$$= \int_0^1 x^2 \cdot k x^2 (1-x^3)\,dx - \left(\frac{9}{14}\right)^2$$

$$= 6\int_0^1 (x^4 - x^7)\,dx - \frac{81}{196}$$

$$= 6\left[\frac{x^5}{5} - \frac{x^8}{8}\right]_0^1 - \frac{81}{196}$$

$$= 6\left[\frac{1}{5} - \frac{1}{8}\right] - \frac{81}{196}$$

$$= 6\left[\frac{8-5}{40}\right] - \frac{81}{196}$$

$$= \frac{18}{40} - \frac{81}{196} = \frac{9}{20} - \frac{81}{196}$$

$$= \frac{9(196) - 81(20)}{20(196)}$$

$$= \frac{1764 - 1620}{3920}$$

$$= \frac{\cancel{144}\,9}{\cancel{3920}\,245} = \frac{9}{245}$$

iv) $P(0 < x < 0.5)$

$$P(0 < x < 0.5) = \int_0^{0.5} 6x^2(1-x^3)\,dx$$

$$= 6\int_0^{0.5} (x^2 - x^5)\,dx$$

$$= 6\left[\frac{x^3}{3} - \frac{x^6}{6}\right]_0^{0.5}$$

$$= 6\left[\frac{(0.5)^3}{3} - \frac{(0.5)^6}{6}\right]$$

$$= 6\left[\frac{0.125}{3} - \frac{0.015625}{6}\right]$$

$$= 6[0.041667 - 0.002604]$$

$$= 6[0.039063]$$

$$P(0 < x < 0.5) = 0.2344$$

Ten unbiased coins are tossed simultaneously. obtain the Probability of a) Exactly 6 heads b) at least 8 heads c) No heads d) Not More that 3 heads.

Soln:- The probability of getting 'r' heads in 'n' tosses of an unbiased coin is given by the Binomial Distribution

$$P(x=r) = {}^{n}C_{r}\, p^{r} q^{n-r}, \quad n=10$$

$P \rightarrow$ Probability of getting a Head $(P) = \frac{1}{2}$

$$\& \; q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$$

i) Exactly 6 heads

$$P(x=6) = {}^{10}C_{6} \left(\tfrac{1}{2}\right)^{6} \left(\tfrac{1}{2}\right)^{10-6}$$

$$= 210 \times \frac{1}{2^{10}}$$

$$= 210 \times \frac{1}{1024}$$

$$P(x=6) = 0.2051$$

ii) Atleast 8 heads

$$P(x \geq 8) = P(x=8) + P(x=9) + P(x=10)$$

$$= {}^{10}C_{8} \left(\tfrac{1}{2}\right)^{8} \left(\tfrac{1}{2}\right)^{10-8} + {}^{10}C_{9} \left(\tfrac{1}{2}\right)^{9} \left(\tfrac{1}{2}\right)^{10-9} + {}^{10}C_{10} \left(\tfrac{1}{2}\right)^{10} \left(\tfrac{1}{2}\right)^{10-10}$$

$= 45 \times \frac{1}{2^{10}} + 10 \times \frac{1}{2^{10}} + \frac{1}{2^{10}} = \frac{45}{1024} + \frac{10}{1024} + \frac{1}{1024}$

$= \frac{1}{1024}[45 + 10 + 1]$

$= \frac{1}{1024}[56] = 0.0547$

iii) No Heads

$P(x=0) = {}^{10}C_0 (1/2)^0 (1/2)^{10-0}$

$= 1 \cdot \frac{1}{2^{10}} = \frac{1}{1024} = 0.0009766$

iv) Not more then 3 Heads:

$P(x \leq 3) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$

$= {}^{10}C_0 (1/2)^0 (1/2)^{10-0} + {}^{10}C_1 (1/2)^1 (1/2)^{10-1} + {}^{10}C_2 (1/2)^2 (1/2)^{10-2} + {}^{10}C_3 (1/2)^3 (1/2)^{10-3}$

$= 1 \cdot \frac{1}{2^{10}} + 10 \cdot \frac{1}{2^{10}} + 45 \cdot \frac{1}{2^{10}} + 120 \cdot \frac{1}{2^{10}}$

$= \frac{1}{2^{10}}[1 + 10 + 45 + 120]$

$= \frac{1}{2^{10}}[176] = \frac{1}{1024}[176]$

$= \frac{176}{1024} = 0.171875$

6. The mean of binomial distribution is 3 and the variance is $\frac{9}{4}$

Find i) $P(x \geq 4)$ ii) $P(1 \leq x < 6)$ iii) $P(x \leq 3)$ iv) $P(4 < x \leq 7)$

Sol^n:- Given Mean of binomial Distribution $= 3$

i.e. $np = 3 \rightarrow (1)$

Variance of binomial Distribution $= \frac{9}{4}$

i.e. $npq = \frac{9}{4} \rightarrow (2)$

$2 \div 1 = \frac{npq}{np} = \frac{9/4}{3} = \frac{9}{4} \times \frac{1}{3} = \frac{3}{4}$

$q = \frac{3}{4}$; $p = 1 - q = 1 - \frac{3}{4} = 1/4$

$p = \frac{1}{4}, q = \frac{3}{4}$

from (1) $np = 3 \Rightarrow n \times \frac{1}{4} = 3 \Rightarrow n = 12$

i) $P(x \geq 4) = 1 - P(x < 4)$

$= 1 - [P(x=0) + P(x=1) + P(x=2) + P(x=3)]$

$= 1 - [{}^{12}C_0 (1/4)^0 [3/4]^{12-0} + {}^{12}C_1 [1/4]^1 [3/4]^{12-1} + {}^{12}C_2 [1/4]^2 [3/4]^{12-2} + {}^{12}C_3 [1/4]^3 [3/4]^{12-3}]$

$= 1 - \left[1 \cdot \frac{3^{12}}{4^{12}} + 12 \cdot \frac{3^{11}}{4^{12}} + 66 \cdot \frac{3^{10}}{4^{12}} + 220 \cdot \frac{3^9}{4^{12}}\right]$

$= 1 - \frac{3^9}{4^{12}} [3^3 + 12[3^2] + 66[3] + 220]$

$= 1 - \frac{19683}{16777216} [27 + 108 + 198 + 220]$

$= 1 - \frac{19683}{16777216} (553)$

$= 1 - \frac{10854699}{16777216}$

$= 1 - 0.6488$

$P(x \geq 4) = 0.3512$

ii) $P(1 \leq x < 6) = P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5)$

$= {}^{12}C_1 (1/4)^1 (3/4)^{12-1} + {}^{12}C_2 (1/4)^2 (3/4)^{12-2} + {}^{12}C_3 (1/4)^3 (3/4)^{12-3} +$

${}^{12}C_4 [1/4]^4 [3/4]^{12-4} + {}^{12}C_5 [1/4]^5 [3/4]^{12-5}$

$$= 12\,\frac{3^{11}}{4^{12}} + 66\,\frac{3^{10}}{4^{12}} + 220\,\frac{3^{9}}{4^{12}} + 495\,\frac{3^{8}}{4^{12}} + 792\,\frac{3^{7}}{4^{12}}$$

$$= \frac{3^{7}}{4^{12}}\left[972 + 1782 + 1980 + 1485 + 792\right]$$

$$= \frac{2187}{16777216}\,(7011)$$

$$P(1 \le x < 6) = 0.9139$$

iii) $P(x \le 3) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$

$$= {}^{12}C_0\,[1/4]^0\,[3/4]^{12-0} + {}^{12}C_1\,[1/4]^1\,[3/4]^{12-1} + {}^{12}C_2\,[1/4]^2\,[3/4]^{12-2} + {}^{12}C_3\,[1/4]^3\,[3/4]^{12-3}$$

$$= 1\,\frac{3^{12}}{4^{12}} + 12\,\frac{3^{11}}{4^{12}} + 66\,\frac{3^{10}}{4^{12}} + 220\,\frac{3^{9}}{4^{12}}$$

$$= \frac{3^{9}}{4^{12}}\left[1(3^3) + 12[3^2] + 66[3] + 220\right]$$

$$= \frac{19683}{16777216}\left[27 + 108 + 198 + 220\right]$$

$$= \frac{19683}{16777216}\,[553]$$

$$P(x \le 3) = 0.6488$$

iv) $P(4 < x \le 7) = P(x=5) + P(x=6) + P(x=7)$

$$= {}^{12}C_5\,(1/4)^5\,[3/4]^{12-5} + {}^{12}C_6\,[1/4]^6\,[3/4]^{12-6} + {}^{12}C_7\,[1/4]^7\,[3/4]^{12-7}$$

$$= 792\,\frac{3^{7}}{4^{12}} + 924\,\frac{3^{6}}{4^{12}} + 792\,\frac{3^{5}}{4^{12}}$$

$$= \frac{3^{5}}{4^{12}}\left[792(3^2) + 924(3) + 792\right]$$

$$= \frac{243}{16777216}\left[7128 + 2772 + 792\right]$$

$= \frac{243}{16777216}[10692]$

$P[4<x\leq 7] = 0.1548$

7. If the Probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the Probability that out of 2000 individuals i) exactly 3 individuals ii) more than 2 individuals iii) None iv) More than one individual suffer a bad reaction

Soln:- Given $p=0.001$ & $n=2000$

$\therefore$ Mean $= \lambda = np = 2000 \times 0.001 = 2$

$\lambda = 2$

$$P(r) = \frac{e^{-\lambda}\lambda^r}{r!} = \frac{e^{-2}\cdot 2^r}{r!} = \frac{2^r}{r!e^2}$$

i) $P(x=3) = \frac{2^3}{3!e^2} = \frac{8}{6(2.718)^2}$

$= \frac{8}{6(7.3875)} = \frac{8}{44.325} = 0.1804$

ii) P(more than 2) $= P(x>2)$

$= 1 - P(x\leq 2)$

$= 1 - [P(0) + P(1) + P(2)]$

$= 1 - \left[\frac{2^0}{0!e^2} + \frac{2^1}{1!e^2} + \frac{2^2}{2!e^2}\right]$

$= 1 - \frac{1}{e^2}[1+2+2]$

$= 1 - \frac{1}{7.3875}[5] = 1 - 0.6768$

$= 0.3233$

iii) $P(\text{none}) = P(0) = \frac{2^0}{e^2 0!} = \frac{1}{e^2} = \frac{1}{7.3875}$

$= 0.1353$

iv) $P(\text{more than one}) = P(r>1) = 1 - P(r \le 1)$

$= 1 - [P(0) + P(1)] = 1 - \left[\frac{1}{e^2} + \frac{2}{e^2}\right]$

$= 1 - \frac{3}{e^2} = 1 - 3(0.1353) = 1 - 0.406$

$= 0.594$

8. a] A car hire firm has two cars which it fires out day by day. The number of demands for a car on each day is distributed as poisson variate with mean 1.5. calculate the proportion of days on which i) neither car is used, and ii) some demand is refused.

Given mean $\lambda = 1.5$

we have $P(r) = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$

i) $P(\text{neither car is used}) = P(0)$

$= \frac{e^{-1.5} \times (1.5)^0}{0!} = \frac{e^{-1.5} \times 1}{1} = e^{-1.5} = 0.2231$

ii) $P(\text{some demand is refused}) = P(r>2)$

$= 1 - P(r \le 2)$

$= 1 - [P(0) + P(1) + P(2)]$

$= 1 - \left[e^{-1.5} + \frac{e^{-1.5}(1.5)^1}{1!} + \frac{e^{-1.5}(1.5)^2}{2!}\right]$

$$+\frac{e^{-5}\cdot 5^8}{8!}+\frac{e^{-5}\cdot 5^9}{9!}+\frac{e^{-5}\cdot 5^{10}}{10!}\Bigg]$$

$$=1-e^{-5}\left[1+5+\frac{25}{2}+\frac{125}{6}+\frac{625}{24}+\frac{3125}{120}+\frac{15625}{720}+\frac{78125}{5040}+\frac{390655}{40320}\right.$$

$$\left.+\frac{1953125}{362880}+\frac{9765625}{3628800}\right]$$

$$=1-0.0067\left[1+5+12.5+20.83+27.166+26.041+21.70+15.50\right.$$

$$\left.+9.688+5.3822+2.69\right]$$

$$=1-0.0067\left[147.4912\right]$$

$$=1-0.9882$$

$$=\underline{\underline{0.0118}}$$

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Assignment-2

① The mean yield for one-are plot is 662 kgs, with a SD of 32 kgs assuming normal distribution how many one-acre plots in a batch of 1000 plots would you expect to have a yield of (a) over 700 kgs b) below 650 kgs c) In b/w 675 kgs and 725 kgs.

Sol: Given

Mean $\mu = 662$ kgs

SD $\sigma = 32$ kgs

Total plots = 1000

a) Yield over 700 kgs

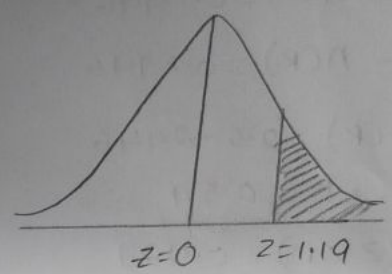
when $x = 700 \Rightarrow z = \frac{x-\mu}{\sigma} = \frac{700-662}{32} = \frac{38}{32} = 1.19$

$P(z > 1.19) = 0.5 - A(z)$

$= 0.5 - A(1.19)$

$= 0.5 - 0.3830$

$= 0.1170$



Batch of 100 plots $= 1000 \times 0.117$

$= 117$

b) below 650 kgs

when $x = 650 \Rightarrow z = \frac{x-\mu}{\sigma} = \frac{650-662}{32} = \frac{-12}{32} = -0.38$

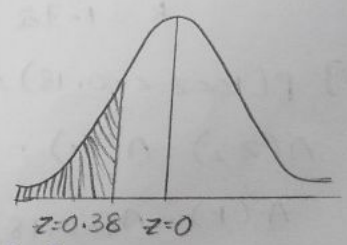
$P(z < -0.38) = 0.5 - A(z)$

$= 0.5 - A(-0.38)$

$= 0.5 - A(0.38)$

$= 0.5 - 0.1480$

$= 0.352$



The batch of 1000 plots $= 1000 \times 0.352 = 352$ plots

c) between 675 kgs and 725 kgs.

when $x = 675 \Rightarrow z_1 = \frac{x-\mu}{\sigma} = \frac{675-662}{32} = \frac{13}{32} = 0.41$

when $x = 725 \Rightarrow z_2 = \frac{x-\mu}{\sigma} = \frac{725-662}{32} = 63/32 = 1.97$

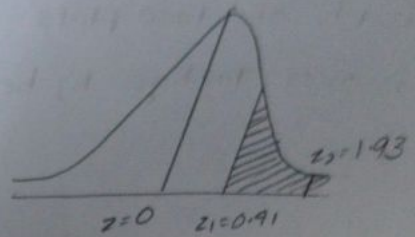
$\therefore P(675 < x < 725) = P(0.41 < x < 1.97)$

$= A(z_2) - A(z_1)$

$= A(1.97) - A(0.41)$

$= 0.4756 - 0.1591$

$= 0.3165$



The batch of 1000 plots $= 1000 \times 0.3165$

$= 316.5$

$= 0.7235 - 0.3238$

$A(K) = 0.3997$

$K = 1.28$ (∵ from table)

Pr (4) :- The marks of SDE subject is distributed normal with mean and standard deviation are 75 and 10 respectively, 20% of the total students are failed in exam. Top 15% of total students got A grade then find (a) what is the minimum to get pass. (b) what is the minimum marks to get grade A.

Sol:- Given mean $\mu = 75$ & $\sigma = 10$

(a) let K = no. of students failed in exam

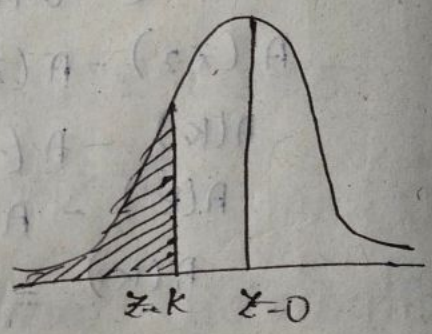
$= 20\% = 0.2$

$P(Z < K) = 0.2$

$0.5 - A(Z) = 0.2$

$A(K) = 0.5 - 0.2$

$A(K) = 0.3$



$K = -0.85$ ∵ from table

minimum pass marks $\Rightarrow Z = \frac{x-\mu}{\sigma}$

$\Rightarrow x = \mu + Z\sigma$

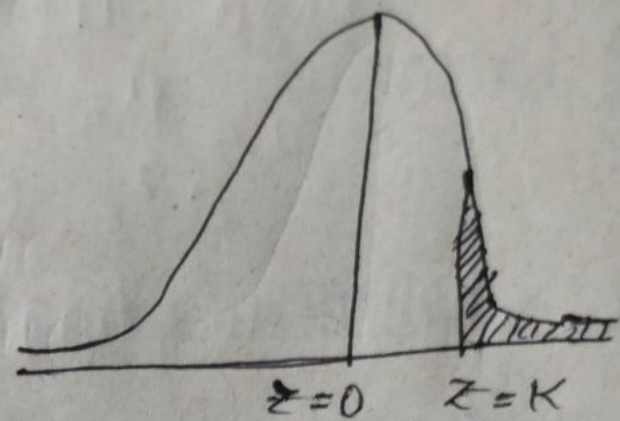
$= 75 + (-0.85)10$

$= 75 - 8.5$

$x = 66.5$

D) let K = no. of students to get $A = 15\%$.

$$A = 0.15$$



$$A(Z > K) = 0.15$$

$$0.5 - A(z) = 0.15$$

$$0.5 - A(K) = 0.15$$

$$A(K) = 0.5 - 0.15$$

$$A(K) = 0.35$$

$$K = 1.04 \quad (\because \text{from table})$$

$$z = \frac{x - \mu}{\sigma} \Rightarrow x = \mu + z\sigma$$

$$x = 75 + 1.04(10)$$

$$x = 75 + 10.4$$

$$x = 85.4$$

Q5 :- what is the effect on standard error, if a sample taken from an infinite population of sample size is increased from 400 to 800. (a)

Sol:- Given standard error of mean

$$S.E = \frac{\sigma}{\sqrt{n}} \quad \text{where } n \rightarrow \text{sample size}$$

for old sample size $n = n_1 = 400$

$$S.E_1 = \frac{\sigma}{\sqrt{400}} = \frac{\sigma}{20} \rightarrow (1)$$

for new sample size $n = n_2 = 800$

$$S.E_2 = \frac{\sigma}{\sqrt{800}} = \frac{\sigma}{20\sqrt{2}} \rightarrow (2)$$

eq(2) ÷ eq(1)

$$\Rightarrow \frac{SE_2}{SE_1} = \frac{\frac{\sigma}{20\sqrt{2}}}{\sigma/20} = \frac{1}{\sqrt{2}}$$

$$S.E_2 = \frac{1}{\sqrt{2}}(SE_1)$$

when sample size is doubled, the sample error is decresed by a factor of $\frac{1}{\sqrt{2}}$.

ie. $SE_2 = 0.707 \ (S.E_1)$

$SE_2 = 70.7\%$ of SE_1

percentage decrease $= 100\% - 70.7\%$

$= 29.3\%$

(b) a random sample of size 100 is taken from an infinite population having the mean $\mu = 76$ and the variance $\sigma^2 = 256$, what will probability that $\bar{x}$ will be

between 75 and 78.

Sol: given $n = 100$,

$\mu = 76$,

$\sigma^2 = 256 \Rightarrow \sigma = \sqrt{256} = 16$

we have to find $P(75 < \bar{x} < 78) = ?$

W.K.T $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

from $\bar{x} = 75 \Rightarrow z = \dfrac{75 - 76}{16/10} = \dfrac{-1}{1.6} = -0.625$

for $\bar{x} = 78 \Rightarrow z = \dfrac{78 - 76}{16/10} = \dfrac{2}{1.6} = 1.25$

$\therefore P(75 < \bar{x} < 78) = P(-0.625 < z < 1.25)$

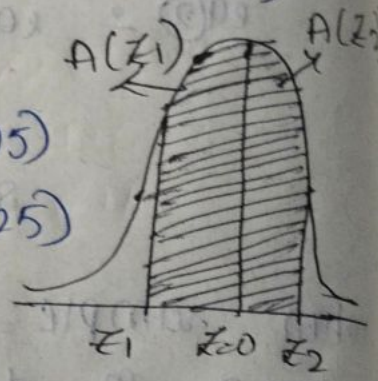
$= A(z_2) + A(z_1)$

$= A(1.25) + A(-0.625)$

$= A(1.25) + A(0.625)$

$= 0.3944 + 0.2357$

$= 0.63$



12-2-26

⑥ The guaranted average life of a certain type of electric bulbs is 1500 hrs with a S.D of 100 hrs, it is desired to sample the output so as to ensure that 95% of bulbs don't fall short the guaranted average by more than 8%

what will be the minimum sample size.

Sol: let 'n' be the size of the sample

the guaranted mean $\mu = 1500$ hrs

we do not want the mean of the sample to be less than 2% of $1500 = \frac{2}{100} \times 1500$

$= 30$ hrs

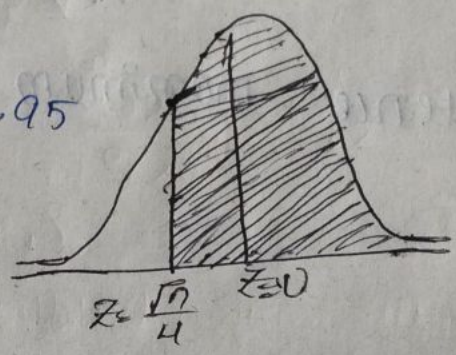
so the sample mean should be $> 1500 - 30 = 1470$

ie. $\bar{x} > 1470$

$$\therefore |z| = \frac{|\bar{x} - \mu|}{\sigma/\sqrt{n}} = \frac{|1470 - 1500|}{120/\sqrt{n}} = \frac{\sqrt{n}}{4}$$

by given condition

$P(\bar{x} \geq 1470) = 95\% = 0.95$



ie. $P(z \geq \frac{\sqrt{n}}{4}) = 0.95$

ie. $0.5 + A(z) = 0.95$

$0.5 + A\left(\frac{\sqrt{n}}{4}\right) = 0.95$

$A\left(\frac{\sqrt{n}}{4}\right) = 0.95 - 0.5$

$A\left(\frac{\sqrt{n}}{4}\right) = 0.45$

$\frac{\sqrt{n}}{4} = 1.65$

$\sqrt{n} = 4 \times 1.65 = 6.6$

$n = (6.6)^2 = 43.56$

$\therefore n \leq 44$.

4(a) What is the maximum error one can expect to make with probability 0.90, when using the mean of a random sample of size $n = 64$, estimate the mean of population with $\sigma^2 = 2.56$?

Sol: Given here $n = 64$,

the probability $= 0.90$

$\sigma^2 = 2.56 \Rightarrow \sigma = 1.6$

confidence limit $= 90\%$

$z_{\alpha/2} = 1.645$

Hence maximum error $= z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

$= 1.645 \times \frac{1.6}{\sqrt{64}}$

$= 0.329$

7(b): It is desired to estimate the mean no. of hours of continuous use until a certain computer will first require repairs. If it can be assumed that $\sigma = 48$ hrs, how large a sample be needed so that one will be able to assert with 90% confidence that the sample mean is off by at most 10 hrs.

Sol: It is given that

maximum error $E = 10$ hrs

$\sigma = 48$ hrs

If $Z_{\alpha/2}$ at 90% $= 1.645$

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{E}\right)^2$$

$$= \left[\frac{1.645 \times 48}{10}\right]^2$$

$\therefore$ The sample size $(n) = 62.3 \cong 62$.

(8) The pulse rate of 50 yoga practitioness decreased on the average by 20.2 beats/minutes with S.D of 3.5

(a) If $\bar{x} = 20.2$ is used as a point estimate of the true average decrease in pulse rate what can we assert with 95% confidence about the maximum error E

(b) construct a 99% confidence interval for the true average decrease in pulse rate.

Sol: Given $n = 50$

$\bar{x} = 20.2$

$SD = 3.5$

(a) maximum error $= Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

$$= 1.96 \times \frac{3.5}{\sqrt{50}}$$

$= 1.96 \times 0.495$ (confidence level

$= 0.97$ 95% at $z = 1.96$)

(b) Given $n = 50$, $\bar{x} = 20.2$, $SD = 3.5$

confidence level = 99%, at $z = 2.58$

confidence interval $= \left(\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right)$

$= 20.2 \pm 2.58 \left(\frac{3.5}{\sqrt{50}}\right)$

$= 20.2 \pm 1.3$

$= (18.9, 21.5)$

$\therefore$ 99% confidence interval $= (18.9, 21.5)$